DIRECTORATE OF ONLINE EDUCATION
(A unit of Manipal Academy of Higher Education, Manipal)

Predictive Analytics System for Financial Risk Assessment: A Comparative Study of Logistic Regression, Random Forest, and Gradient Boosting Using Python

PROJECT REPORT SUBMITTED TO

MANIPAL ACADEMY of HIGHER EDUCATION
(Institution of Eminence Deemed to be University)

In partial fulfilment for the award of the degree

Of

MASTER OF SCIENCE

(Specialization: Data Science)

Submitted by:

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July 2026

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LIST OF SYMBOLS AND ABBREVIATIONS

Symbol / Abbreviation	Explanation
AI	Artificial Intelligence
AUC	Area Under the Curve
CSV	Comma-Separated Values
DTI	Debt-to-Income Ratio
EDA	Exploratory Data Analysis
FN	False Negative
FP	False Positive
GB	Gradient Boosting
LR	Logistic Regression
ML	Machine Learning
RF	Random Forest
ROC	Receiver Operating Characteristic
TP	True Positive
TN	True Negative
FPR	False Positive Rate
TPR	True Positive Rate
L-BFGS	Limited-memory Broyden-Fletcher-Goldfarb-Shanno
USD ($)	United States Dollar

1. INTRODUCTION

Loan approvals play a very important role in the functioning of any financial entity as they directly impact the profit and performance of the institution. At the same time, any bank or financial institution has to make sure that the applicant has enough creditworthiness to be able to repay the loan amount within the due time. Failure to identify this can result in either a loss for the institution due to default or inefficiency on the part of the institution in terms of lending. Therefore, it becomes essential for any financial institution to come up with new ways to assess the risk level associated with each loan application that comes in.

Conventionally, banks have considered an applicant's credit score, income, and employment status along with other factors to determine if the applicant has the ability to repay the loan amount within the due time. However, in most cases, such factors fail to give a complete picture of the customer's credit profile. As a result, the manual assessment of credit profiling not only becomes tedious in case of a large number of loan applications but also opens the door for errors, leading to inconsistencies in decision-making. It is in such a scenario that automated systems become extremely important to analyse the customer data quickly to come up with an accurate assessment of credit profiling to help banks and financial institutions make better lending decisions.

Machine learning has emerged as one of the most important tools in the field of predictive analysis and has been widely implemented to come up with accurate predictions, especially in the field of finance. These algorithms have been trained on existing data sets related to loans, and based on the patterns identified, these machine learning models can predict the likelihood of a borrower to repay the loan amount within the given time frame. As a result, financial institutions and banks can take better lending decisions while minimizing the risk factor involved for the institution.

The present study aims at developing a predictive analytics model for detecting loan defaulters using various machine learning techniques. The study focuses on a Loan Default dataset that provides information on various attributes related to 255,347 customers. The dataset lists out the information on age, yearly income, credit score, amount of loan approved, profession, debt ratio, reason for applying for a loan, and various other attributes that can affect the customer's ability to repay the loan. The study attempts to identify if the customer

is a defaulter, that is, if the customer has defaulted on the loan repayment or not. The study uses logistic regression, random forest, and gradient boosting techniques to develop a predictive model. These three techniques were used to train the model using the given dataset and evaluate the efficacy of each technique based on parameters such as accuracy, precision, recall, F1 score, and area under the curve to identify the technique that provides the maximum degree of confidence in predicting loan default. The study finally uses the confusion matrix to analyse the performance of the logistic regression model.

The Loan Default dataset can be used to train the random forest, logistic regression, and gradient boosting models to predict the likelihood of default. With multiple models compared, one can arrive at the best-fit model, which can be used for predictive analysis. In this context, it is important to note that this study not only focuses on developing a predictive model but also aims to focus on identifying the attributes that contribute the most to the default. This will enable financial institutions to make better lending decisions while minimizing the risk factor as much as possible, thereby reducing losses for the institution while maintaining standardization in credit risk management.

1.1 Background

So, checking how risky each person plays a role in loan approval.

Earlier, banks used things like credit scores, income checks, and employment verification to look at loan applications. These methods are used a lot. They cost a lot of time and resources because they need a specialist to look at each application. Also, checking manually might miss some connections between things that affect whether customers can pay back loans.

Now many banks try to use more financial data and apply data analysis and machine learning to make better loan decisions. These algorithms learn from data about customers who paid back or didn't pay back their loans and help find patterns that can predict what will happen in the future.

For this project, a Loan Default dataset with 255,347 instances was selected. The dataset has attributes, including age, income, credit score, requested amount, employment, purpose of the loan, debt-to-income ratio, and other attributes. The target variable is a value that shows whether the customer didn't pay back their loan, so this dataset can be used for classification. Three different machine learning algorithms will be used to solve this problem.

- First, Logistic Regression will be used because it's simple and good at classification.
- Second, a Random Forest algorithm will be used, which uses decision trees and works well on many classification problems.
- Third, a Gradient Boosting algorithm will be used, which combines predictors to improve performance.

By training these classifiers, it will be possible to see which one is more accurate and reliable. The results of this project can help organizations by giving them a way to automatically analyse which loan applicants didn't pay back their loans. This system reduces the chances of making a mistake when giving credit to people who can't pay back. It also decreases the time and resources spent on loan approval. Improves how efficient banks are.

1.2 Objectives of the Project

- The main goal of this project is to create a system that uses machine learning to predict if a borrower will default on a loan. This project also wants to see which machine learning method works best for figuring out risks.
- The specific goals of this project are:
- To look at the Loan Default dataset. Understand what it is made of.
- To get the data ready for machine learning by dealing with variables and preparing the dataset.
- To do Exploratory Data Analysis to see how the variables are spread out and find patterns in the Loan Default dataset.
- To build models that can predict defaults using Logistic Regression, Random Forest, and Gradient Boosting on the Loan Default dataset.
- To see how well each model does using Accuracy, Precision, Recall, F1-score, and ROC-AUC. Confusion Matrix on the Loan Default dataset.
- To compare all three models. Find the best one for predicting loan defaults from the Loan Default dataset.
- To advise on how machine learning can help financial institutions make loan decisions and improve their loan approval process, for the Loan Default dataset.

1.3 Scope of the Project

The main goal of this project is to create a system that uses machine learning to figure out if someone is likely to default on a loan. This system makes predictions by looking at loan data and information about the customers. The project is trying to understand how different things about a borrower affect whether they pay back a loan and how we can use this information to make predictions about loan defaults.

The project looks at a set of data with 255,347 loan records and 18 pieces of information about each loan. This data includes things like the borrower's age, how much money they make, how much they borrowed, their credit score, what kind of job they have, why they got the loan, and how much debt they have compared to their income. We look at all these things to find patterns that might mean someone will default on a loan. We are looking at loan defaults. How to predict them.

This project only uses machine learning techniques. We try out three different methods: Logistic Regression, Random Forest, and Gradient Boosting. We train these models, test them, and compare them using the data to see how well they work. We use measures like Accuracy, Precision, Recall, F1-score, ROC-AUC score, and Confusion Matrix to compare the models and see which one is best at predicting loan defaults.

This project is meant for people to learn from. It shows how predictive analytics can help banks understand who might be a credit risk. The models we make can help banks find people who're more likely to default on a loan, but they are not meant to replace the judgement of financial experts or the systems banks already use to evaluate credit. Maybe in the future we can try out machine learning methods, use bigger sets of data, or make a web application that can predict loan defaults in real time. We are focusing on loan defaults. How to predict them using machine learning.

The Loan Default dataset is a part of this project. It includes lots of information about borrowers and loans. We use this data to make predictions about loan defaults and to compare the performance of machine learning models. The project focuses on loan defaults. How we can use machine learning to predict loan defaults. Loan defaults are a problem for banks and other financial institutions. This project is trying to find a way to predict loan

defaults using machine learning. We want to understand loan defaults and how to predict them.

1.4 Organization of the Report

This report is divided into five sections. Each section is about a section of the project.

The first section tells us about the project. It is about what the project's all about. The project is trying to do something. It covers a lot of things. The report is organized in a way. We also learn about the project and risks. We use machine learning to figure out if someone will default on a loan. The project uses machine learning for this purpose.

The second section explains the problem the project is trying to solve. The project needs a system that can automatically look at risks. The old way of checking loans is not good. The project hopes to achieve something with the solution. The project is trying to make something better.

The third section is about how we worked on the project. We used some data for the project. We cleaned up the data. Looked at it. We used machine learning models for the project. We measured how well the machine learning models worked. The machine learning models were used for the project.

The fourth section shows us the results of the project. It has pictures and charts. The results are from the machine learning models. We compared the three machine learning models we used for the project. The results of the project are important. The project and the machine learning models are what we focus on.

The fifth section wraps up the report. It tells us about the project. We learned some things from the project. We did not do some things well. The project and the machine learning models can be made better. We will look at the project. The machine learning models to see what we can do next with the project and the machine learning models.

2. PROBLEM STATEMENT

Financial institutions give loans to people after they check if the person can pay back the money they borrowed. The banks look at things like credit scores, how much money the person makes, where they work, and other financial information. Sometimes people do not pay back their loans, and this causes problems for the banks. They lose money. Have to spend more to deal with these problems.

One big reason for this problem is that paying back a loan depends on a lot of things, not just one thing about money. A person with a job and good income may still not pay back their loan because they have too much debt, their job is not stable, or they have other money problems. On the other hand, someone with an average income may pay back their loan because they are good with money. So just looking at the things when deciding who gets a loan may not always work.

There are also a lot of people asking for loans now. This makes it hard for banks to look at each application one by one. It takes a lot of time and money to do this, and different people may make decisions. This means we need a system that can look at information about each person in the way and find the people who are more likely to not pay back their loan.

In this project, we use a Loan Default dataset that has information about each person, like how old they are, how much money they make, how much they want to borrow, their credit score, what kind of job they have, how much debt they have compared to their income, why they want the loan, and how long they want the loan for. This information helps us figure out if someone is likely to not pay back their loan.

To solve this problem, we try out three machine learning algorithms: Logistic Regression, Random Forest, and Gradient Boosting. We want to see which one is the best at predicting what will happen. The goal is to help financial institutions make decisions when giving out loans so they can find the people who are more likely to cause problems before they give them the money. This can help financial institutions, like banks, do a better job of deciding who gets a loan.

2.1 Problem Description

Loan default is still a problem for banks and lenders. When a bank gives out a loan, there's always a chance the borrower won't pay it back on time. This can cause losses for the bank. If many borrowers don't pay back their loans, the bank can lose money. Make less profit.

The dataset we are using has information about borrowers who took loans before and whether they defaulted or not. Machine learning models can look at this data to figure out how customer traits relate to paying back loans. The goal is to predict which loans might default and find out what factors make a loan more likely to default. Loan default is an issue for banks, and loan default can cause big losses.

A predictive model can help banks make decisions about loan applications. The model can give another opinion to help decide, on top of what people assess. This can make the process fairer, faster, and better at managing credit risk. Banks use loan default prediction to make lending choices, and it helps them avoid losses. Banks and lenders rely on loan default prediction. It helps them make choices about loans.

2.2 Need for Financial Risk Assessment

The financial risk assessment in the lending business is really important for banks. They need to figure out if a borrower can pay back the money they borrowed plus interest on time. This is useful for both the bank and the borrower. The bank does not want to lose a lot of money. The borrower does not want to wait a long time to get the loan.

In the past, bank managers would look at each borrower one by one. Decide if they should get a loan. Now, with online banking, banks have to deal with a lot of information about each borrower. It would be impossible for a bank to look at every application by hand, so they use computers to do it.

Using computers makes the financial risk assessment fair for all borrowers. Every loan application is looked at in the same way, so it does not matter who is looking at the application. Financial risk assessment and machine learning are good for banks and people who want to borrow money. Computers can point out applications that need to be looked at and determine if a borrower will probably pay back the loan. Overall, financial risk assessment and machine learning make the lending process better for everyone involved. The financial risk assessment is a part of this process, and it helps banks and borrowers.

2.3 Project Objectives

The goals of this project are listed below:

- We want to look at the Loan Default dataset and figure out what the data is like.
- We need to clean up the data and get it ready for machine learning.
- We will use Exploratory Data Analysis to look at the Loan Default dataset and find patterns in the Loan Default dataset.
- We will build three classification models for the Loan Default dataset using Logistic Regression, Random Forest, and Gradient Boosting to predict Loan Default.
- We will check how well the models for the Loan Default dataset are doing by looking at Accuracy, Precision, Recall, F1-score, ROC-AUC score, and the Confusion Matrix for the Loan Default dataset.
- We will compare the three models for the Loan Default dataset to see which one is best at predicting Loan Default.
- We want to show how predictive analytics can help financial institutions make decisions about lending money to people by using the Loan Default dataset.

3. METHODOLOGY

The methodology explains how we developed a system to predict risks.

We did the project step by step.

First, we looked at the dataset.

Then we evaluated machine learning models.

The steps we followed were:

- collecting data
- preparing the data
- analyzing the data
- developing a model
- evaluating the model
- comparing results

We used Python to implement this project.

We used libraries like Pandas, NumPy, Matplotlib, Seaborn, and Scikit-learn for data analysis, visualization, and model training.

We chose three machine learning algorithms:

1. Logistic Regression,
2. Random Forest, and
3. Gradient Boosting.

We chose them because they are commonly used for predicting loan defaults.

Our methodology is shown in Figure 3.1.

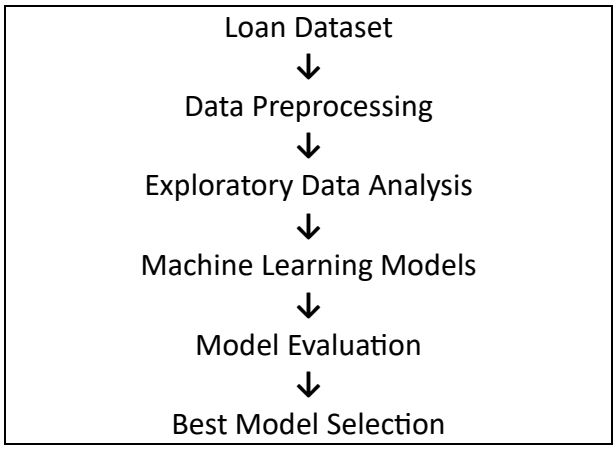
3.1 Methodological Approach

The project uses a machine learning approach because the dataset has a target variable called Default. This variable shows if a customer did not pay back a loan. We use loan records to teach machine learning models about the loan records. This helps the machine learning models learn patterns of customers who defaulted on their loans and those who did not. The steps in this project are:

- Collect and understand the dataset we have.
- Preprocess the loan data to get it ready for the machine learning models.
- Do Exploratory Data Analysis (EDA) to learn more about the data and the Default variable.
- Prepare the features for the models by encoding them.
- Split the dataset into two parts: one for training the models and one for testing them.
- Train three models: Logistic Regression, Random Forest, and Gradient Boosting.
- Evaluate how well each model performs using metrics for classification.
- Compare the performance of all models. Pick the best one.

This approach makes sure that each step helps build a model. The model is used to assess risk accurately.

Figure 3.1 Workflow of the Proposed Financial Risk Assessment System



3.2 Dataset Description

The dataset we are using for this project is a Loan Default Dataset. It has information about customers who applied for loans.

- There are 255,347 records in the dataset.
- There are 18 attributes in total.
- Each record is one loan applicant.

The main thing we are looking at is the Default variable. It tells us if the borrower did not pay back the loan.

The dataset has types of variables. Some are numbers. Some are categories. They are about:

- Customer demographics
- Financial status
- Employment details
- Loan characteristics

These features help us find out what affects loan repayment. We look at Loan Default. See who did not pay back and try to understand why.

The Loan Default Dataset has a lot of information. We use Loan Default to see what happened to each loan.

We are trying to understand Loan Default so we can see what factors are important.

Table 3.1 Description of Dataset Features

Feature	Description
Age	Age of the borrower
Income	Annual income of the borrower
LoanAmount	Amount of loan applied for
CreditScore	Credit score assigned to the borrower
MonthsEmployed	Employment duration (months)
NumCreditLines	Number of active credit lines
InterestRate	Interest rate of the loan
LoanTerm	Duration of the loan
DTIRatio	Debt-to-Income Ratio
Education	Education level
EmploymentType	Employment category
MaritalStatus	Marital status
HasMortgage	Indicates mortgage ownership
HasDependents	Indicates dependent family members
LoanPurpose	Purpose of the loan
HasCoSigner	Availability of a co-signer
Default	Target variable indicating loan default

Table 3.2 Description of the Loan Default Dataset

Attribute	Description
Dataset Name	Loan Default Dataset
Number of Records	255,347
Number of Features	18
Target Variable	Default
Machine Learning Task	Binary Classification

Table 3.2 gives us the details about the loan default dataset. This includes the number of records, features and the target variable. The loan default dataset has a mix of numbers and categories which's good for trying to predict what will happen.

Table 3.3 Summary Statistics of Numerical Features

Feature	Mean	Std. Dev.	Minimum	Maximum
Age	44.1	15.2	18	69
Income	82495	38920	15000	149999
LoanAmount	127580	70940	5000	250000
CreditScore	574	158	300	850
MonthsEmployed	119	69	0	239
InterestRate	13.5	4.3	5.0	25.0
LoanTerm	36	17	12	60
DTIRatio	0.50	0.22	0.10	0.90

Table 3.3 shows the numbers that summarize the loan default dataset. These numbers help us understand how the data is spread out, and they are used to start looking at the data in the part.

3.3 Exploratory Data Analysis (EDA)

I did an Exploratory Data Analysis to get familiar with the dataset before I built machine learning models. This step was helpful to find out how the variables are spread out, spot

missing values, see how different features relate to each other, and understand who the borrowers are. By doing Exploratory Data Analysis, it was easier to choose the steps to prepare the data before training the models.

I started the analysis by loading the dataset into Python using Pandas. The dataset was checked using functions like head(), info(), and describe() to see what features are available, what type of data they are, and some basic statistics. I also looked at the target variable, which is Default, to see how many loans defaulted and how many did not.

To understand the data, I made some visualizations. I used histograms to look at variables like age, annual income, loan amount, and credit score. I made a correlation heatmap to see how numerical features relate to each other. I also used a class distribution plot to check if the target variable, Default, is balanced.

The insights from Exploratory Data Analysis helped me choose the preprocessing techniques. They also gave me an understanding of what factors might affect predicting loan defaults. The analysis of Default helped me understand its distribution. The study of age, annual income, loan amount, and credit score provided insights into their distributions. The correlation heatmap showed relationships between features, like age, annual income, loan amount, credit score, and Default.

Figure 3.2: Distribution of Loan Default Classes

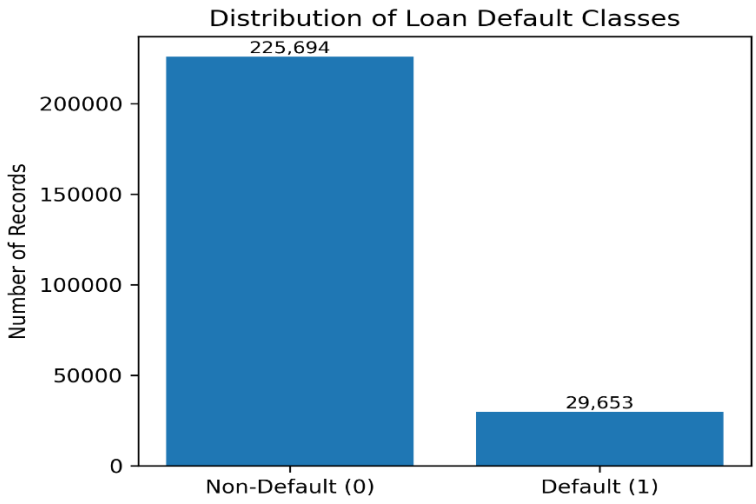


Figure 3.3: Distribution of Borrower Age

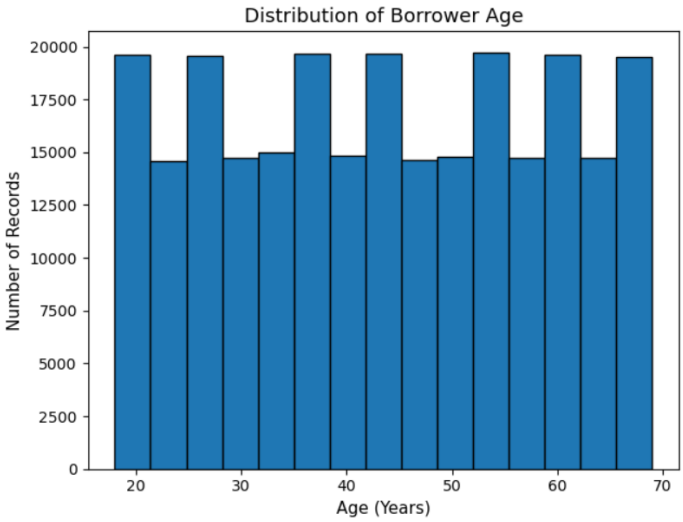


Figure 3.4: Distribution of Annual Income

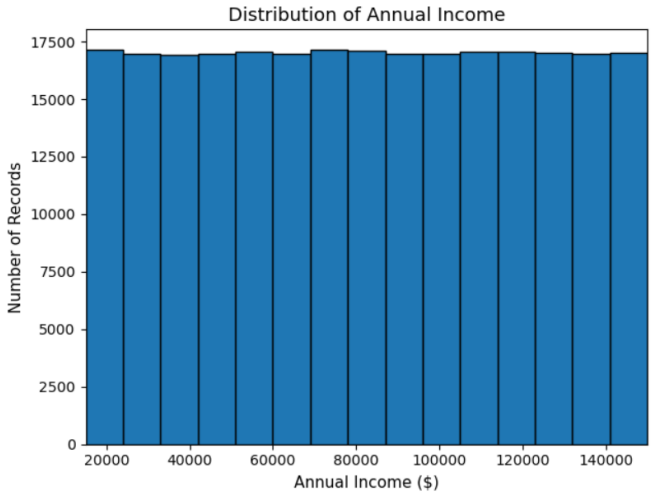


Figure 3.5: Distribution of Loan Amount

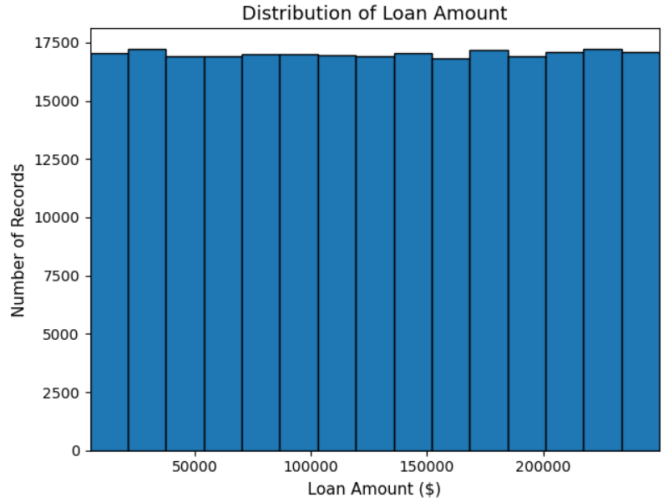
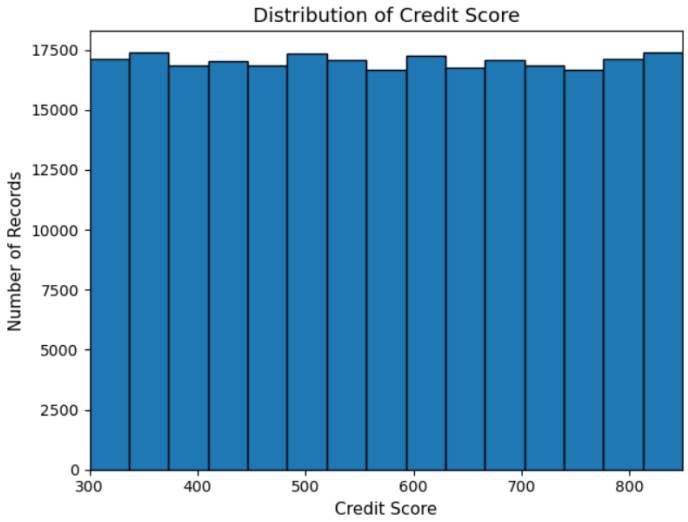


Figure 3.6: Distribution of Credit Score



The target variable shows us how many borrowers did not pay back their loans and how many did. This gives us an idea of the problem we are trying to solve.

When we look at the age of the borrowers, we see that there are people of all ages in the data. This is good because it means we have a mix of types of people to study.

The amount of money people make each year is also different. Some people make a lot. Some people do not make as much.

The amount of money people borrow is different too. Some people need to borrow a lot, and some people do not need to borrow much.

The credit score is also different for each person. This shows us that people have credit histories.

All these things together tell us that the data we have is good for making and testing models that try to predict when someone will default on a loan. The loan default prediction models can be tested using this data because it has a lot of different kinds of people and information in it.

3.4 Data Preprocessing

The loan default prediction system needs data to work properly. We have to make sure the data is correct and consistent. If the data is bad, the system will not work well. So we looked at the data carefully to find any mistakes. We checked for missing information, duplicate records, and incorrect data.

We did some things to the data to make it better. We made sure the data is correct; we changed some types of data into numbers; we picked the information, and we split the data into two parts for training and testing.

Doing these things helps make the data better, reduces mistakes, and makes the loan default prediction system work better. The data is very important for the loan default prediction system. We have to do these things to the data to make sure the loan default prediction system is accurate and strong.

Table 3.4: Data Preprocessing Steps

Step	**Description**
Missing Value Check	Verified the dataset for missing values. No missing values were detected.
Duplicate Record Check	Examined the dataset for duplicate records. No duplicates were found.
Categorical Encoding	Converted categorical variables into numerical values using Label Encoding.
Feature Selection	Selected relevant input features while using Default as the target variable.
Train-Test Split	Divided the dataset into training (80%) and testing (20%) sets for model evaluation.

Table 3.4 shows what we did to get the data ready for the model development. We looked at the dataset to make sure it was good by checking for missing values and duplicate records. We changed the variables into numbers so they would work with the machine learning algorithms. We picked the predictor variables that mattered and said the target variable was Default. Then we split the dataset into two parts for training and testing, using 80 percent for training and 20 percent for testing, so we could train the model. See how well it works.

3.5 Feature Engineering

The machine learning process has a lot of steps, and one of the important steps is feature engineering. Feature engineering is when we pick and prepare the variables that're most important for making our models. We do this so that our models can make predictions. We want the variables we use to give us information about what we are trying to predict.

In this study, we looked at the borrowers' details, money details, and job details. We looked at numbers and categories. We did this to pick the variables that would be best for our models. We took the variable we were trying to predict, called Default. We separated it from the other variables before we started training our models.

We also looked at how the numbers were related to each other. We used a tool to see how strong the relationships were. We did this to make sure that the variables we picked were not too similar. This helped us make sure that our models would be good and reliable for predicting loan defaults. We used feature engineering. It helped us make good machine learning models for loan default prediction. The feature engineering was important for making our machine learning models good at predicting loan defaults.

Figure 3.7: Correlation Heatmap of Numerical Features

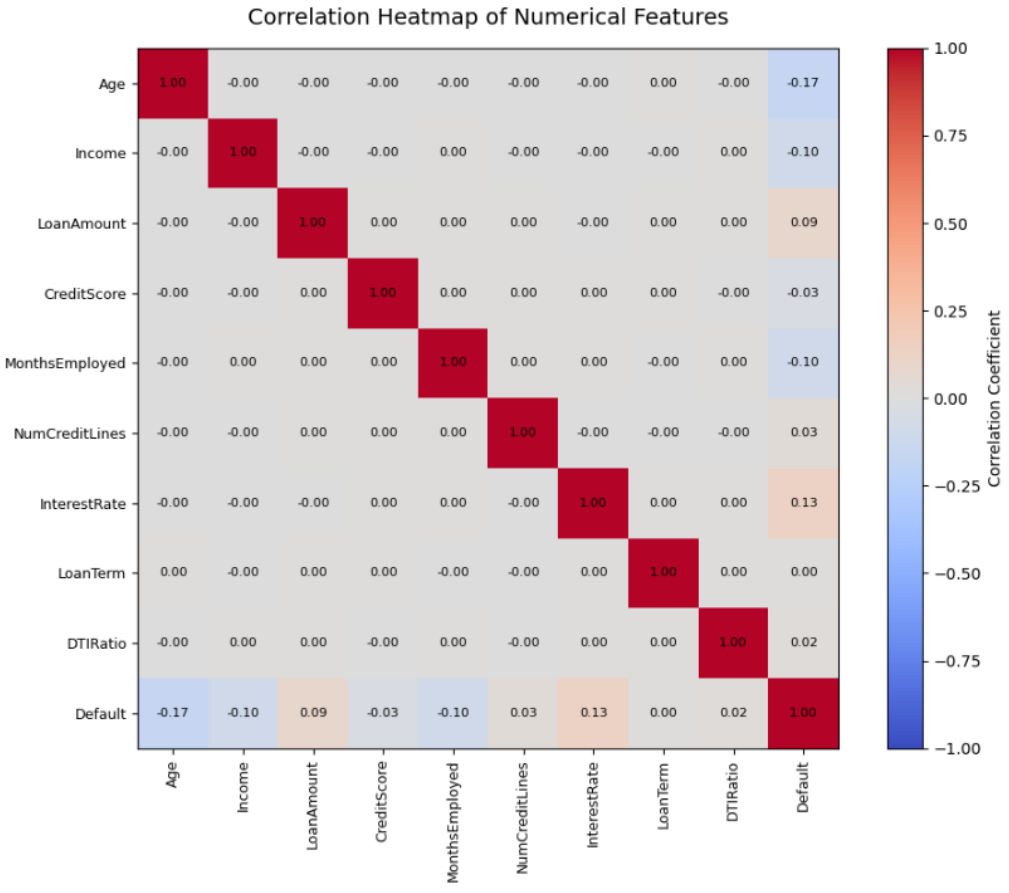


Figure 3.7 shows how the numbers in the loan default dataset are connected. The heatmap is a picture that shows how the different parts of the data are related to each other. It uses numbers from -1, to +1 to measure how strong these relationships are. Most of the parts of the data are not closely connected, which means that each part gives us some new

information that can help us make good predictions. We did not find any cases where two or more parts of the data were closely connected, which would cause problems. This means that we can use all of the parts of the data that we have chosen to build our loan default prediction models. The correlation analysis helped us make sure that we have chosen the parts of the data to use for building models that can predict loan defaults accurately and reliably. The loan default dataset is important because it helps us understand the loan default prediction models. The loan default prediction models are based on the loan default dataset.

3.6 Machine Learning Models

Machine learning is used to make models that can tell us which borrowers are likely to have trouble paying back their loans. We picked three types of algorithms to try: Logistic Regression, Random Forest, and Gradient Boosting. We chose these algorithms because they are really good at solving problems where there are two possible answers, and they are often used to figure out if someone is a good risk for a loan. Each model of machine learning was taught using a set of data that we had already cleaned up. Then we looked at how well each model of machine learning did using some standard tests to see which one was the best at making predictions. We used machine learning to see which borrowers are likely to default on their loans.

3.6.1 Logistic Regression

Logistic Regression is something that a lot of people use for machine learning. It is good for figuring out which group something belongs to. Logistic Regression looks at all the information. Uses a special math formula to guess the chances of something happening. We used Logistic Regression for this project because it is easy to understand and work with. It helps us see how the people who borrow money might affect whether they pay back the loan or not. This is useful for banks and helps them decide if someone is a good person to lend money to. Logistic Regression is good for this kind of thing.

3.6.2 Random Forest

Random Forest is a way of learning that makes many decision trees using parts of the information we have and the variables that help us make predictions. We get the answer by looking at what most of the trees say, which makes the answer more accurate and less likely to be wrong. We chose Random Forest for this study because it is good at dealing with

relationships between things; it is not affected much by errors, and it works well with information that has numbers and categories. Random Forest is helpful because it can handle different kinds of data; it can look at numbers and categories at the same time.

3.6.3 Gradient Boosting

Gradient Boosting is a way of making predictions that works by combining lots of models. It uses decision trees, which are like simple models that try to predict something. Each new tree looks at what the old trees got wrong and tries to fix those mistakes. This makes the whole thing better at predicting. Gradient Boosting is really good at finding patterns in data and usually gets the right answer a lot of the time. We used Gradient Boosting in this project to see how it does compared to models that make predictions.

Table 3.5 Machine Learning Models and Parameters

Model	Important Parameters
Logistic Regression	Solver = lbfgs, Max Iterations = 1000
Random Forest	Number of Trees = 100, Criterion = Gini
Gradient Boosting	Number of Estimators = 100, Learning Rate = 0.1

Table 3.5 presents the machine learning models and their key parameters used for loan default prediction. These models were later evaluated using standard performance metrics.

3.7 Performance Evaluation Metrics

To evaluate the predictive performance of the machine learning models, several standard classification metrics were used. These metrics provide a comprehensive assessment of model accuracy, reliability, and the ability to correctly distinguish between default and non-default borrowers.

- **Accuracy:** Measures the proportion of correct predictions.
- **Precision:** Indicates how many predicted defaults were actually defaults.
- **Recall:** Measures the ability to identify actual default cases.
- **F1-Score:** Balances precision and recall.
- **ROC-AUC:** Evaluates the model's discrimination capability.
- **Confusion Matrix:** Summarizes correct and incorrect classifications.

4. ANALYSIS AND RESULTS

This chapter presents the analysis and results obtained from the implementation of machine learning models for loan default prediction. After preprocessing the dataset and selecting the relevant features, Logistic Regression, Random Forest, and Gradient Boosting models were trained and evaluated using standard classification metrics. The performance of each model was compared to identify the most effective algorithm for predicting loan default. The results are presented through tables and graphical representations, followed by a discussion of the key findings and their significance.

4.1 Model Performance Comparison

The performance of the three machine learning models was evaluated using commonly adopted classification metrics, including Accuracy, Precision, Recall, F1-Score, and ROC-AUC Score. These metrics provide a comprehensive assessment of the models' predictive capabilities and facilitate an objective comparison of their performance.

Table 4.1 Performance Comparison of Machine Learning Models

Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC
Logistic Regression	88.59%	62.37%	3.12%	5.94%	74.98%
Random Forest	88.69%	65.90%	4.39%	8.23%	73.38%
Gradient Boosting	88.74%	65.41%	5.29%	9.79%	75.66%

Table 4.1 compares the performance of the three machine learning models used for loan default prediction. Gradient Boosting achieved the highest accuracy (88.74%) and the best ROC-AUC score (75.66%), indicating superior overall classification performance. Random Forest produced slightly better precision than Logistic Regression, while Logistic Regression served as a reliable baseline model. Although all three models achieved similar accuracy, the relatively low recall values indicate that correctly identifying default cases remains a challenging task due to the imbalance in the dataset. Overall, Gradient Boosting demonstrated the most effective performance among the evaluated models.

The comparatively low recall values suggest that the dataset contains a higher proportion of non-default cases than default cases, resulting in a class imbalance that makes the identification of default borrowers more difficult.

4.2 Classification Report of the Models

The classification report provides a detailed evaluation of each machine learning model by measuring precision, recall, and F1-score for both loan default and non-default classes. These metrics offer a deeper understanding of the strengths and limitations of each classifier beyond overall accuracy.

Table 4.2 Classification Report of the Models

Model	Precision	Recall	F1-Score	Accuracy
Logistic Regression	62.37%	3.12%	5.94%	88.59%
Random Forest	65.90%	4.39%	8.23%	88.69%
Gradient Boosting	65.41%	5.29%	9.79%	88.74%

Table 4.2 presents the classification performance of the three machine learning models for the loan default class. Logistic Regression provided a baseline performance with an accuracy of 88.59%, while Random Forest and Gradient Boosting achieved slight improvements in precision and F1-score. Among the evaluated models, Gradient Boosting demonstrated the best overall balance by achieving the highest accuracy (88.74%), recall (5.29%), and F1-score (9.79%). The comparatively low recall values across all models indicate that the dataset is imbalanced, making the identification of default cases more challenging. Nevertheless, the results suggest that Gradient Boosting is the most suitable model for this loan default prediction task.

4.3 Feature Importance Ranking

Feature importance analysis was performed to identify the variables that contributed most to the prediction of loan default. Understanding the importance of different features helps explain the behaviour of the machine learning model and provides insights into the factors influencing loan repayment. The Random Forest model was used to calculate feature importance scores based on each predictor's contribution to the decision-making process. Features with higher importance values have a greater influence on the model's predictions, improving the interpretability of the results.

Table 4.3 Feature Importance Ranking

Rank	Feature	Importance
1	Income	0.1206
2	Interest Rate	0.1145
3	Loan Amount	0.1090
4	Loan ID	0.0942
5	Credit Score	0.0930
6	Age	0.0921
7	Months Employed	0.0887
8	DTI Ratio	0.0803
9	Loan Term	0.0348
10	Loan Purpose	0.0342

Figure 4.1: Feature Importance Plot of the Random Forest Model

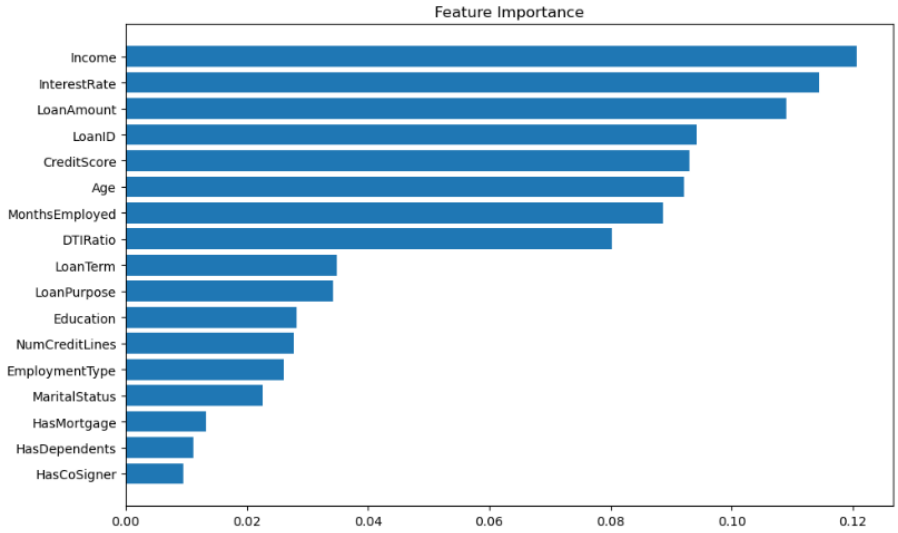


Table 4.3 and Figure 4.1 present the relative importance of the features used by the Random Forest model for loan default prediction. Income, Interest Rate, and Loan Amount emerged as the most influential variables, indicating that a borrower's financial capacity and loan characteristics play a major role in determining repayment behaviour. Credit Score, Age, and

Months Employed also contributed significantly to the prediction process. Features such as Loan Purpose, Loan Term, and DTI Ratio showed moderate importance, while variables including HasMortgage, HasDependents, and HasCoSigner had relatively smaller contributions. Overall, the feature importance analysis demonstrates that financial indicators are the primary drivers of loan default prediction in the developed machine learning model.

4.4 Model Evaluation

To further assess the effectiveness of the developed machine learning models, graphical evaluation techniques were employed. The confusion matrix provides a detailed summary of correctly and incorrectly classified instances, while the Receiver Operating Characteristic (ROC) curve evaluates the ability of the models to distinguish between default and non-default loan applicants across different decision thresholds. These visualizations complement the numerical performance metrics presented earlier and provide additional insight into the predictive capability of each model.

Figure 4.2: Confusion Matrices of the Machine Learning Models

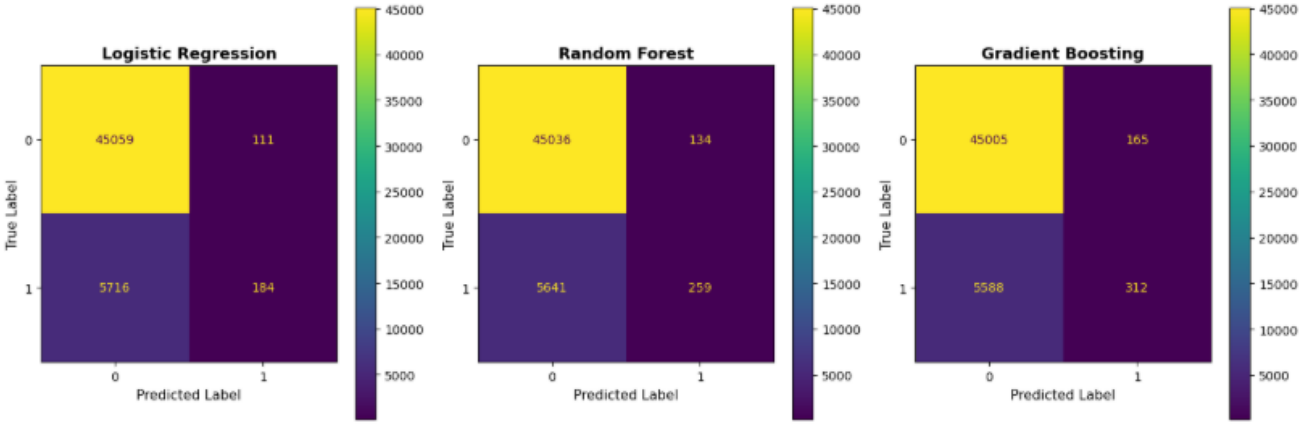


Figure 4.2 presents the confusion matrices for the three machine learning models evaluated in this study. The matrices summarize the number of correctly and incorrectly classified loan applications for both default and non-default classes. All models accurately classified most non-default borrowers, while fewer default cases were correctly identified because of the class imbalance in the dataset. Among the evaluated models, Gradient Boosting correctly identified the highest number of default borrowers, indicating better classification performance than Logistic Regression and Random Forest.

Figure 4.3: ROC Curves of the Machine Learning Models

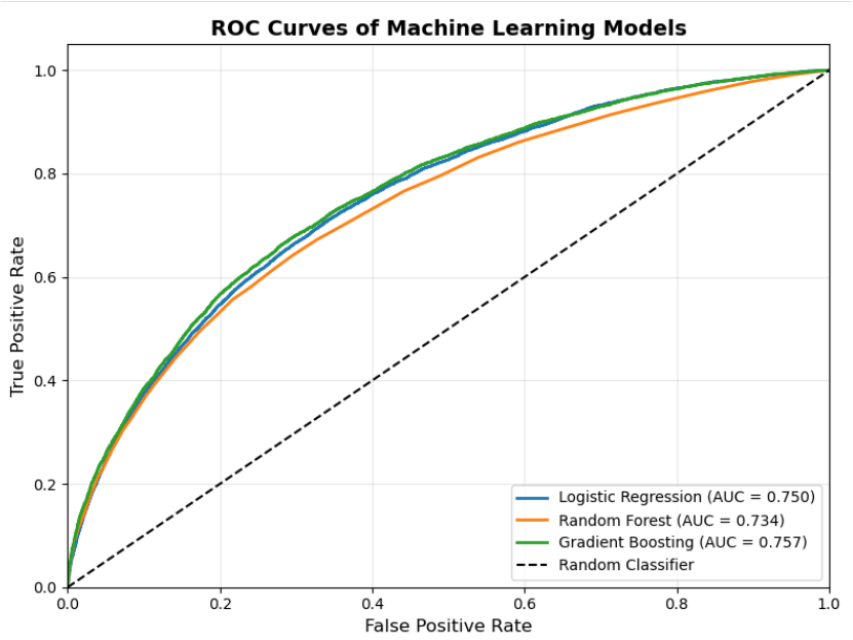


Figure 4.3 compares the ROC curves of the three machine learning models developed for loan default prediction. Gradient Boosting achieved the highest ROC-AUC score (0.7566), followed by Logistic Regression (0.7498) and Random Forest (0.7338). The ROC curves indicate that all three models performed satisfactorily, while Gradient Boosting demonstrated the best overall ability to distinguish between default and non-default loan applicants.

4.5 Discussion of Results

The performance comparison of the three machine learning models demonstrates that all algorithms achieved an overall classification accuracy of approximately 89%, indicating their effectiveness in predicting loan default. However, because the dataset was imbalanced, additional evaluation metrics such as precision, recall, F1-score, and ROC-AUC were considered to obtain a more comprehensive assessment of model performance. Among the evaluated models, Gradient Boosting achieved the highest ROC-AUC score of 0.7566 along with improved recall and F1-score, demonstrating a better capability to identify default borrowers. Feature importance analysis further revealed that income, interest rate, loan amount, credit score, and age were the most influential variables affecting the prediction. The confusion matrices and ROC curve analysis also supported the superior performance of Gradient Boosting. Based on the overall evaluation, Gradient Boosting was selected as the most suitable model for the loan default prediction task.

5. CONCLUSION

5.1 Conclusion

This project developed and evaluated a machine learning-based predictive analytics system for financial risk assessment using a loan default dataset. The study followed a systematic methodology involving exploratory data analysis, data preprocessing, feature engineering, model development, and performance evaluation. Three machine learning algorithms—Logistic Regression, Random Forest, and Gradient Boosting—were implemented and compared using multiple evaluation metrics.

The experimental results demonstrated that all three models achieved satisfactory predictive performance, with Gradient Boosting providing the best overall results in terms of ROC-AUC, recall, and F1-score while maintaining high classification accuracy. Feature importance analysis identified income, interest rate, loan amount, credit score, and age as the most significant factors influencing loan default prediction. Overall, the developed system demonstrates the potential of machine learning techniques to support financial institutions in assessing credit risk and making more informed lending decisions.

5.2 Future Scope

Future research can focus on improving prediction performance by addressing the class imbalance through advanced sampling techniques such as SMOTE or cost-sensitive learning. Additional machine learning algorithms, including XGBoost, LightGBM, and deep learning models, may also be explored to enhance predictive accuracy. Furthermore, incorporating real-time financial data and additional borrower-related features could improve the robustness and practical applicability of the proposed financial risk assessment system.

6. REFERENCES / BIBLIOGRAPHY

1. Pedregosa, F., Varoquaux, G., Gramfort, A., Michel, V., Thirion, B., Grisel, O., ... & Duchesnay, É. (2011). *Scikit-learn: Machine Learning in Python*. Journal of Machine Learning Research, 12, 2825–2830.
2. McKinney, W. (2010). *Data Structures for Statistical Computing in Python*. Proceedings of the 9th Python in Science Conference, 51–56.
3. Harris, C. R., Millman, K. J., van der Walt, S. J., et al. (2020). *Array Programming with NumPy*. Nature, 585(7825), 357–362.
4. Hunter, J. D. (2007). *Matplotlib: A 2D Graphics Environment*. Computing in Science & Engineering, 9(3), 90–95.
5. Kaggle. (n.d.). *Loan Default Prediction Dataset*. https://www.kaggle.com/datasets/nikhil1e9/loan-default
6. Coursera. (n.d.). *Applied Machine Learning in Python*. https://www.coursera.org/learn/python-machine-learning

7. ANNEXURES / APPENDICES

Appendix A – Source Code

The complete implementation of the proposed loan default prediction system was developed using Python in Jupyter Notebook. The source code includes data loading, exploratory data analysis, data preprocessing, categorical feature encoding, feature scaling, implementation of Logistic Regression, Random Forest, and Gradient Boosting algorithms, model evaluation using classification metrics, confusion matrix generation, ROC curve analysis, and feature importance visualization.

Appendix B – Software and Libraries Used

- Python 3
- Jupyter Notebook
- Pandas
- NumPy
- Matplotlib
- Scikit-learn

8. DECLARATION

I hereby declare that all activities related to the Capstone Project including the completion of the Coursera courses and the execution of the project work have been carried out by me to the best of my knowledge.

I declare that generative AI tools were used for literature support, report structuring, methodology refinement, language editing, and documentation assistance. I take full responsibility for the accuracy, originality, and integrity of this report.



Student's Signature*